

Take a Look Around – The Impact of Decoupling Gaze and Travel-direction in Seated and Ground-based Virtual Reality Utilizing Torso-directed Steering

Daniel Zielasko*
Human-Computer Interaction
University of Trier

Yuen C. Law†
School of Computing
Costa Rica Institute of
Technology

Benjamin Weyers*
Human-Computer Interaction
University of Trier



Figure 1: **A seated user from above.** From left to right, visualized are the three different direction components of the investigated steering methods: gaze-directed, torso-directed, and virtual-body-directed. The yellow arrow indicates the direction of travel, which in the left case is given by the head-direction, in the middle case by a tracker mounted to the user's torso, and in the right case by a virtual arrow, which is displayed above the user's head and is controlled by either head rotation or a joystick.

ABSTRACT

Leaning (the upper body) has several times been shown to be a suitable virtual travel technique when being seated; in both, flying as well as ground-based scenarios. The direction of the steering method most commonly used is gaze/head-directed. However, this does not allow to inspect the environment independently from the direction of movement. The change to torso-directed steering allows for the latter and additionally does not take anything from the natural character of the leaning metaphor. We empirically investigated the impact of this freedom in a ground-based scenario and complemented the conditions with a virtual body-directed method and then crossed all with device-based control conditions. In the conducted study ($n = 25$), we found the torso-directed methods objectively performed the best (traveled distance, completion time & number of collisions), and found torso-directed leaning subjectively rated the most usable one.

Index Terms: Human-centered computing—Virtual reality; Human-centered computing—Empirical studies in interaction design

1 INTRODUCTION

Being seated while consuming VR can have diverse reasons and could be dictated by the surrounding real-world environment [23], it can be beneficial for long term usage and ergonomics, necessary due to impairment, to match the simulated real-world activity, or just because of laziness and convenience. In this scenario, leaning (the upper body) has several times been shown to be a suitable virtual travel technique; in both, flying [21, 24] as well as ground-based scenarios [12, 15]. The direction of the steering method most commonly

used is gaze/view-directed, where the actual implementation most of the time is realized with the users' head being tracked and not their eyes. For the remainder of this work, we will treat gaze/view-directed and head-directed as synonyms following the common practice [6, 14, 19, 21]. In both cases, however, gaze-directed steering does not allow to inspect the environment independently from the direction of movement. Giving the travel direction by the direction of leaning is a solution, but moving the upper body outside of the sagittal plane and additionally independently from the head can be tiring over time, which we have subjectively observed in various experiments in the past [24]. Furthermore, this adds lateral movement, so-called *strafing*, which can feel unnatural and is suspected of having a negative impact on cybersickness [5].

Thus, an alternative is to use torso-directed steering, which does not take anything from the natural character of the leaning metaphor [14] and might be used over more extended periods, when seated in a swivel chair and having enough room to turn around freely. However, using torso-directed steering in a swivel chair suggests wireless display technology and additional tracker, both of which only just recently became easily affordable regarding technical expanse and price. This is one possible reason why research on this technique is still sparse (see Section 2).

Now having the option to move the head around freely, one question remains: Is the (potential) additional methodical and technical complexity worth the effort? Therefore, in this work, we investigated if a head that is decoupled from travel can have a positive impact on task performance, movement characteristics, and user experience, in a seated and ground-based scenario. To achieve this, we compared a torso-directed and a virtual body-directed (see Section 3.3 & 3.6) with a gaze-directed leaning method in a relative motion experiment [3]. All three methods were also implemented and evaluated in combination with a gamepad to add a base-line and filter potential *leaning-only* effects (see Figure 2).

*e-mail: {zielasko, weyers}@uni-trier.de

†e-mail: ylaw@itcr.ac.cr



Figure 2: **Experimental Setup.** In half of the conditions, leaning (left) was used to control velocity, in the second half a gamepad (right). In all conditions, the subjects sat in a swivel chair and wore an HTC Vive Pro extended with a wireless adapter as well as an additional torso tracker, which was mounted to the chest using a strap for action cameras.

2 RELATED WORK

The ability to look around while steering through an immersive virtual environment (IVE) is rarely investigated, even when not restricted to seated scenarios. Bowman et al. [3] compared pointing-directed and gaze-directed steering, which differed in this point, in a relative motion experiment and found pointing to perform better. However, as a follow-up, they compared steering techniques (gaze, pointing, torso) in a series of experiments but did not find significant differences [4]. Suma et al. [19] extended the pool of methods with walking metaphors and found real walking outperforming the steering methods in a ground-based, divided attention task, which had a focus on following a target in the IVE. However, they also did not find a difference between the steering methods. Finally, Christou et al. [6] found a pointing-directed outperforming a gaze-directed method in a ground-based wayfinding task. In summary, there seems to be a tendency towards pointing being the more efficient steering interface compared to moving in gaze direction. However, neither are the results unambiguous nor is the reason.

When taking an explicit view on torso-directed travel methods, the two already mentioned evaluations [4, 19], to the best of our knowledge, are only extended by von Kapri et al. [20]. They found mixed results regarding cybersickness and traveled distance when compared to two base-line pointing methods in a flying coin collection task. However, the tested method is more a *posture-directed* steering method rather than a pure torso-directed one. Zielasko et al. [21] also left out torso-directed methods because of the technical effort for potential users and compared different walking-in-place and gaze-directed steering metaphors in a flying search and maneuver task, with the user being seated. However, as Suma et al. [19] they also stated that tracking the torso can have possibilities that should be investigated in more detail. Finally, there is a series of work [7, 15, 16] pointing out the positive impact of full-body rotations in seated and standing scenarios, which should also be of central interest when moving torso-directed.

With this work, we aim at closing this gap by evaluating gaze-directed against torso-directed steering in a combined relative motion and spatial orientation task. Due to the necessity of a pointing device that would occupy the user's hand, we decided not to include a pointer-directed method. However, we did include a virtual

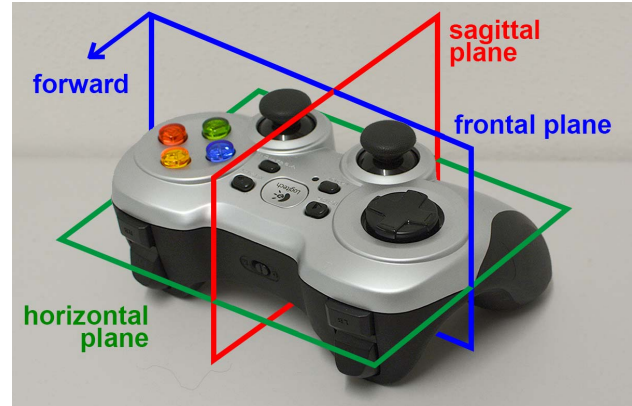


Figure 3: **Gamepad Anatomy.** As there is no uniform standard for the naming of gamepad axes, we are utilizing the anatomical naming conventions through the course of this work.

body-directed metaphor (see Section 3.3 & 3.6) instead. As the velocity control component, we chose a leaning metaphor as it is also usable hands-free and seated, feels natural, and potentially increases self-motion perception [12]. Nevertheless, especially the pure performance of leaning methods over different experiments is very mixed [10, 11, 15, 21], and the results of Guy et al. [8] can be interpreted to use uncorrelated body parts for rotation and translation instead. This led us to the decision to include also joystick-based control conditions. Therefore, in the following evaluation, gaze, torso, and virtual-body-directed steering are additionally implemented and evaluated using a standard gamepad (see Section 3).

3 METHODS

For the outlined investigation, we implemented three different steering methods (gaze vs. torso vs. virtual-body-directed), with two methods of velocity control (joystick vs. leaning) each, resulting in six travel techniques in total. The last two steering methods allow for looking around while moving, from which the torso-directed derives fast from thinking about the requirements and is often mentioned in literature [14].

The virtual-body-directed methods for us originated from different starting points for the two velocity control conditions independently and later formed a common metaphor, the virtual body/torso. In the case of the joystick (gamepad) condition, it was a consequence of bringing the known first-person controls of a classic 3D video game to VR, extended by the possibility to freely look around. A one-to-one translation obviously would lead to gaze-directed steering; however, this does not transport the feeling of the known controls. Differently, in the case of the leaning condition, the metaphor originates from related work [13, 21]. Here this metaphor, also referred to as scrolling [13], is usually used when it is not possible to physically turn the head or body completely, and thus head-yaw induced virtual rotation is introduced.

Thus, both virtual body metaphors also offer virtual rotation. Furthermore, both need a visualization of the virtual body to keep the users informed about their current virtual rotation, namely a green arrow above the users' head (see Figure 4, right). However, depending on the application context, various other virtual bodies are possible, e.g., an avatar, when it is visible enough, a space ship, or a desk [24]. Important to note is that while being hosted under the same metaphor, the two virtual body-directed conditions feel entirely different, and thus we expected that they also perform differently (see the hypotheses in Section 4.6).

For the leaning conditions, a static leaning, i.e., a direct velocity control, was used as this was reported to have a positive impact

on the perception of self-motion [12]. All travel methods were restricted to ground-level. Furthermore, the maximum forward and backward velocity was set to $3m/s = 10.8km/h$ for all conditions, which corresponds to jogging. The maximum angular velocity for virtual rotations of the yaw-axis was set to $100^\circ/s$. In the leaning conditions, to start moving, an angle of 45° around a per user initialized resting position had to be exceeded. For all parameter mappings, linear transfer functions were used. The parameters were chosen in an informal prestudy, independently from the actual study tasks. All implementation details can also be found and reused in the complete sources (including models) available on GitHub¹. A short video showing all methods is available here². In the following, a description of the technical implementation for every method is given.

3.1 Gaze-directed Joystick

In this method, the forward vector of the worn HMD (see Figure 1), projected on the ground was used to determine the direction of travel. The relative tilt of the left analog stick in a gamepad's sagittal plane (see Figure 3) was used to determine velocity, where tilting backward translated to negative velocity and vice versa.

3.2 Torso-directed Joystick

In this method, the forward vector of the worn torso tracker (see Figure 1), projected on the ground was used to determine the direction of travel. The relative tilt of the left analog stick in a gamepad's sagittal plane was used to determine velocity, where tilting backward translated to negative velocity and vice versa.

3.3 Virtual-body-directed Joystick

In this method, a virtual body visualized as a green arrow above the user's head determined the direction of travel (see Figure 4 & 1, right). The arrow (virtual body) could be rotated, using the tilt of the right analog stick in a gamepad's frontal plane (see Figure 3), where a tilt to the left was translated in a counter-clockwise rotation of the virtual body and vice versa. Rotation of the virtual body thus also always simultaneously led to a rotation of the user's view, as long as the user is not actively compensating for this by turning their head in the opposite direction. As the viewpoint was always translated along the direction of the arrow, the users were able to look into another direction as they were actually moving in. Finally, the relative tilt of the left analog stick in a gamepad's sagittal plane was used to determine velocity, where tilting backward translated to negative velocity and vice versa.

3.4 Gaze-directed Leaning

In this method, the forward vector of the worn HMD (see Figure 1), projected on the ground was used to determine the direction of travel. The tilt of the upper body given by the angle between the worn torso tracker's current and initialized forward vector in the users' sagittal plane was used to determine velocity, where tilting back, translated to negative velocity and vice versa.

3.5 Torso-directed Leaning

In this method, the forward vector of the worn torso tracker (see Figure 1), projected on the ground was used to determine the direction of travel. The tilt of the upper body given by the angle between the worn torso tracker's current and initialized forward vector in the users' sagittal plane was used to determine velocity, where tilting back, translated to negative velocity and vice versa.

¹<https://github.com/EagleSixty6/VR-TorsoSteering>

²<https://www.youtube.com/watch?v=96Tb6vjb5-8>

3.6 Virtual-body-directed Leaning

In this method, a virtual body visualized as a green arrow above the user's head determined the direction of travel (see Figure 4, right). The arrow (virtual body) could be rotated by exceeding the angle between the arrow and the camera's forward vector in the transversal plane over more than 45° , i.e., the users had to turn their heads or bodies away from the direction the arrow is pointing in to virtually rotate into this direction. As the viewpoint was always translated along the direction of the arrow, the users were able to look into another direction (up to 45°) as they were actually moving in. The tilt of the upper body given through the angle between the worn torso tracker's current and initialized forward vector in the users' sagittal plane was used to determine velocity, where tilting back, translated to negative velocity and vice versa.

4 EVALUATION

To investigate the influence of the proposed travel methods (see Section 3), which can be categorized through the factors travel-direction (gaze, torso and virtual body) and velocity control (leaning and joystick), on the performance (time, distance, collisions) and usability criteria we conducted a formal within-subjects user study.

4.1 Apparatus

During the whole experimental part of the study, except for potential breaks, the participants sat in a freely rotatable swivel chair (see Figure 2) in the middle of an office room with a linoleum floor. The virtual environment (see Section 4.2) was projected through an HTC Vive Pro that was extended with a wireless adapter. The $2.7m \times 2.7m$ tracking space was given by two HTC Vive Lighthouses 1.0. During all conditions, the participants wore an additional position tracker on their lower chest, which was mounted on a chest strap for action cameras. In the joystick conditions, a Logitech Wireless Gamepad F710 was used for input (see Figure 3).

4.2 Virtual Environment

The virtual environment (VE), as well as the program logic, was created using Unity 2019.3 and the plugin SteamVR. For the VE, six courses (levels) were randomly chosen out of procedural generated non-branching street mazes (see Figure 5). The mazes are built by placing pairs of opposite facing small roman temples in a row, leaving gaps of $8m$, which formed side streets that are closed with walls in a depth of about $7m$ (see Figure 4 & 6). The distance of the temple pairs, and thus the width of the main street was set to $5m$. Every 3rd-7th pair of buildings was followed by a 90° left or right turn. The levels end in a dead-end that included a yellow finishing line at chest height. All levels had 64 side streets/gaps, as these were essential for the main task (see Section 4.3). This led to an average level length of about $734m$. The temples could not be entered as collisions with the environment were recorded but avoided. However, moving physically into a wall was not prohibited, but this did not occur during the experiment.

Next to the levels, there was a training stage where every travel method could be tested (see Figure 4, left). The dimensions of this training environment were roughly $35m \times 12m$. In the middle, there were four pillars placed to motivate participants to train steering and maneuvering around them.

4.3 Tasks

The participants performed one level for each of the travel conditions. During a level, they were given two tasks to solve.

The primary task was a relative motion task, namely to follow the course of the street and collect all coins that were placed in the side streets. The participants did not have to move to the target coins to collect them but only needed to look at them for $50ms$. The coin disappearing, and playing a *pling* sound indicated a successful collection. The coin collection itself was not designed to be



Figure 4: **The virtual environment.** To the left, the training level is shown. It contains some pillars to train to steer. The other depictions show snapshots from the study levels from the ego-perspective, also including some collectible coins. To the right, a green arrow above the subjects' head indicates the current direction of travel (virtual-body-directed conditions).

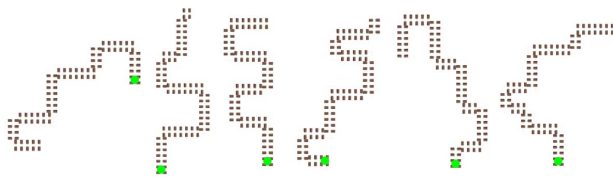


Figure 5: The course of the six used levels. In each case, the green point marks the starting point.

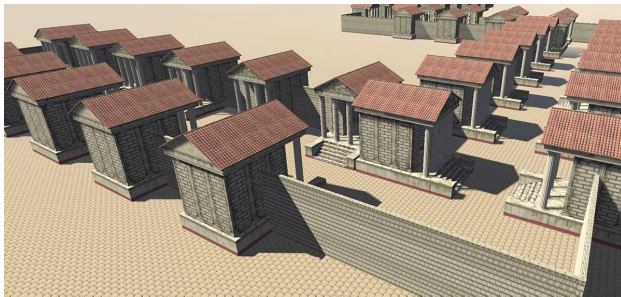


Figure 6: View over a level.

challenging; it just forced the participants to look into every side street. With some skill and agility, it was possible to collect the coins neither without changing the direction nor the speed of travel, by only moving the head, as long as the current travel method provided this.

Every level contained 67 coins. They were randomly distributed once per level between the street pockets using roughly the following distribution: The chance for a pocket to contain no coin was about 2/7, one coin 1/2, and two coins 3/14. Whenever a street pocket contained one or two coins, they were randomly placed close to the closing wall, to the left, right, or middle. Targeting the coins was not realized using eye-tracking but the head direction, such as in the gaze-directed travel methods. To help in doing so, the participants were provided with a small 3D crosshair in the form of a small black sphere.

The secondary task was a spatial orientation task. The participants were instructed before the start of the first level that they should try to keep their orientation. They were also told that this is asked for at the end of every level, where they have to point back to the levels starting point.

The participants were explicitly instructed that collecting all coins was their primary objective above following the spatial orientation

task and being fast.

4.4 Procedure

The user study had the following procedure. Only healthy subjects with normal or corrected-to-normal vision were invited to participate in the study. Every participant was first guided and seated in a room separate from the actual apparatus. Here they were informed about the procedure of the study in written form. Then, they received and had to sign an informed consent. Finally, they were asked to answer a short questionnaire regarding demographics and prior experiences pseudonymously.

Next, participants were guided to the actual place of the experiment. Here they were helped to dress with the chest strap, which the torso tracker was attached to (see Section 4.1), store the wireless adapter's battery pack, take a seat, and put on the HMD. As soon as everything was adjusted, comfortable, and working, they were teleported to the training stage. Here the first travel method was orally explained by the experimenter and simultaneously tried by the participant. Whenever the used travel method involved leaning, the resting position was calibrated. Participants were told at this point that the resting position could be recalibrated at any time. The travel methods were assigned to the subjects in a balanced order using Latin square design. Participants that canceled the experiment were replaced, and overhangs were assigned a distinct random order.

When the participants felt prepared, they were orally instructed about their tasks (see Section 4.3) and teleported to the starting point of the first level. As soon as the subjects crossed the finishing line, travel was disabled, and they were asked to point into the direction of their starting point. Afterward, they were teleported into a recreation room, where they had to answer different questions about their recent experiences:

- *How do you feel, when compared to the beginning of the last level?* – [worse, same, better]
- *How present ('really there') did you feel during the last level?* – [absolutely (-2), ..., not at all (2)]
- *Was the travel method physically demanding to use?* – [absolutely (-2), ..., not at all (2)]
- Eight items from the short version of the user experience questionnaire [18]

Due to restrictions in time, it was not possible to fill out a complete presence and SSQ [9] questionnaire per condition. However, we decided to at least get a partial insight with one overall question each, such as discussed in [1, 22]. Similarly, we used the short version (UEQ-S) [18] of the user experience questionnaire (UEQ) to measure the pragmatic and hedonic quality of the interfaces subjectively.

After having completed these, they could optionally take off the HMD and have a break. Then, they were assigned the next travel method and started the procedure from the training stage again. This happened until they had experienced all six methods. The levels were always assigned in the same order (see Figure 5, from left to right). The whole procedure took on average 80min per participant, of which about 70min were spent in VR.

The study procedure, including data acquisition and handling, was submitted to and approved by the ethical board of the University of Trier.

4.5 Participants

The 25 participants (24 male and 1 female, ages $M = 26.0$ years, $SD = 3.6$ years) were recruited at the campus of the University of Trier. They partially participated voluntarily or were compensated with bonus points for a lecture if applicable. 5 of the participants, including the only female, canceled the study because of nausea. All but one of them stopped already in the first third of the experiment, one first-time user even after 1 minute, with no visible accumulation on the currently used travel technique. 3 of the 5 had no prior experience with VR. The study data of all 5 was not considered for the analysis. Of the remaining participants, 75% reported having prior experiences with 3D video games, such as first-person shooters, 80% with a gamepad, only 45% with VR, and surprisingly already 20% with leaning as an input method for travel.

4.6 Hypotheses

Regarding the core performance criteria, we formulate the following hypotheses.

(H1) We expect the trials performed with methods that allow freely looking around while moving, to take less time than those that do not, because of them requiring the users to stop their movement.

(H2) We explicitly exclude *virtual-body-directed leaning* from the set of methods that support **freely** looking around, as we expect that this ability is significantly interfered by the virtual rotation that is also triggered by turning the head or body.

(H3) Furthermore, we expect that the gaze-directed methods require more distance to finish a level, as subjects will *accidentally* move into the side streets while inspecting these.

(H4) We expect the torso-directed methods to have the highest degree in *environmental presence* and controllability, which results in fewer collisions with the environment. **(H4.1)** Furthermore, we predict that torso-directed leaning induces the least number of environmental collisions, as it is the method with the highest degree of physical body involvement. **(H4.2)** As a direct consequence of H3, we expect the gaze-directed methods to have the highest number of collisions. Previous work partially has shown that joystick-based travel can be more precise [10] and more helpful in maneuvering on a small scale, which would lead to partially contradicting hypotheses. However, we do not think that we observe this, as the tasks do not have a focus on these kinds of movements.

(H5) We expect that the more significant the impact of body cues are, the better the spatial orientation would be. Performing the level turns using physical rotation (torso-directed) should increase the overall ability to maintain orientation [2, 7, 15, 16]. Additionally, it is imaginable that leaning supports traveled distance estimation and thus, in sum, would lead in participants performing the best in the spatial orientation task when using torso-directed leaning.

(H6) Finally, we expect torso-directed leaning to provide the best *look & feel* or hedonic quality because interfaces with physical involvement have shown in the past to increase the overall user experience [14].

5 RESULTS

The objective measures were analyzed using repeated-measures ANOVA. Whenever Mauchly's test indicated that the assumption of

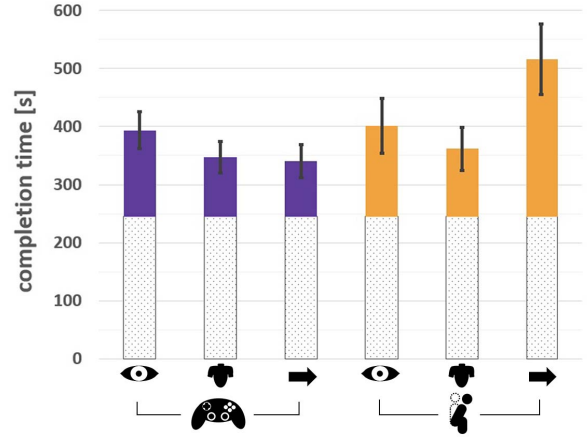


Figure 7: Mean level completion times in seconds for all conditions: $\{\text{gamepad, leaning}\} \times \{\text{gaze, torso, virtual-body-directed}\}$. The white dotted part of the bars indicates the mean minimal completion time (244.9s). Error bars show the 95% confidence intervals. Refer to Section 5.1 for the significance analyses.

sphericity had been violated, the degrees of freedom were corrected using Greenhouse-Geisser estimates. For the post-hoc analysis, a Bonferroni correction was applied. Throughout the paper, all results are reported at a significance level of .05 and non-significant trends at a level of .1.

First, we have a look at the number of missed coins, to control whether the participants followed their instructions that collecting the coins is the primary task. We find 8029 of 8040 coins being collected through the course of the study (error rate 0.14%). Only one participant missed more than one coin per condition once, namely two. We conclude that no participant performed significantly better in any other performance criteria by willingly or accidentally leaving out coins. Thus we can investigate all the following performance criteria without further considering the errors made.

5.1 Completion Time

The mean level completion times are given in Table 1 and depicted in Figure 7. The figure does also visualize the relation of the methodical difference to the minimal completion time, which was determined by completing every level at maximum velocity without collecting any coin. A one-way ANOVA revealed a significant effect of the travel method on the completion time, $F(2.7, 51.1) = 28.7$, $p < .001$.

The post-hoc test showed that virtual-body-directed leaning was significantly slower than all other methods **(H2)**, gaze-directed joystick $t(19) = 5.5$, $p < .001$, torso-directed joystick $t(19) = 7.7$, $p < .001$, virtual-body-directed joystick $t(19) = 8.6$, $p < .001$, gaze-directed leaning $t(19) = 8.6$, $p < .001$, and torso-direct leaning $t(19) = 7.2$, $p < .001$.

Furthermore, the gaze-directed joystick method was significantly slower than the torso-directed joystick $t(19) = 5.6$, $p < .001$, and the virtual-body-directed joystick $t(19) = 5.3$, $p < .001$. For also being slower than torso-directed leaning, there was only a non-significant trend $t(19) = 1.8$, $p = .082$, and finally, there was no difference found to gaze-directed leaning $t(19) = -.43$, $p = .666$.

The gaze-directed leaning method was significantly slower than the torso-directed joystick $t(19) = 2.9$, $p = .008$, and the virtual-body-directed joystick $t(19) = 3.7$, $p = .001$. For being slower than torso-directed leaning, there was again a non-significant trend $t(19) = 2.1$, $p = .053$.

Table 1: Mean (M) and standard deviation (SD) of all core measures for all conditions: {gamepad, leaning} $\times$ {gaze, torso, virtual-body-directed}.

	gamepad						leaning					
	gaze		torso		virtual-body-directed		gaze		torso		virtual-body-directed	
	M	SD	M	SD	M	SD	M	SD	M	SD	M	SD
completion time [s]	393.9	70.5	346.1	60.4	340.5	63.1	401.3	105.7	361.8	82.9	516.0	133.0
traveled distance [m]	787.1	55.5	743.2	26.2	748.5	34.5	838.2	66.1	747.1	20.7	792.5	48.6
# collisions	7.6	8.9	1.4	2.0	1.9	3.1	13.6	10.8	0.7	0.9	7.1	8.2
orientation error [°]	67.3	52.1	55.1	51.3	51.0	42.6	71.7	45.0	52.8	41.2	89.1	54.2
UEQ-S	.41	.87	.94	.54	.38	.66	.42	.67	1.05	.54	-.29	.64
UEQ-S pragmatic	.80	1.02	1.31	.96	1.49	.74	.29	.99	1.13	1.09	-1.19	.87
UEQ-S hedonic	.01	1.18	.56	.72	-.74	.98	.55	.79	.97	.86	.60	.80

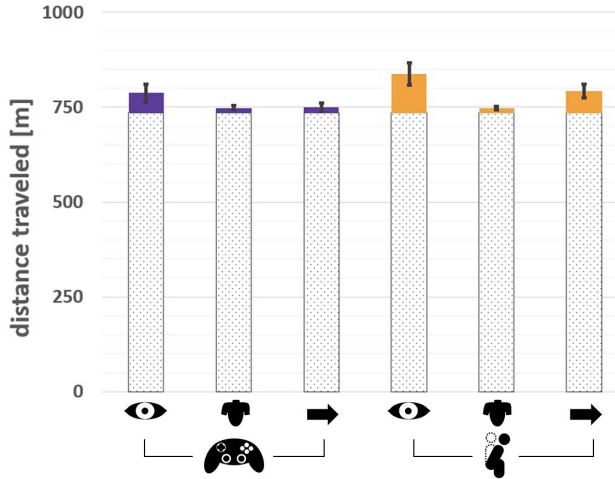


Figure 8: Mean traveled distance forward in meter for all conditions: {gamepad, leaning} $\times$ {gaze, torso, virtual-body-directed}. The white dotted part of the bars indicates the mean minimal necessary distance (734m). Error bars show the 95% confidence intervals. Refer to Section 5.2 for the significance analyses.

There were no statistically significant differences found between the two torso methods and virtual-body-directed leaning. In summary, the methods that support the user in looking around freely were significantly or at least showed non-statistical trends for being faster than those who do not (H1).

5.2 Distance Traveled

The mean distances traveled are given in Table 1 and depicted in Figure 8. The figure does also visualizes the relation of the methodical difference to the minimal distance, which was determined by completing every level at the shortest path to the finishing line.

We hypothesized a main effect of *travel direction* on the traveled distance (H3) and thus performed a two-way ANOVA in this case. The statistical analysis revealed an interaction effect between the factors *travel direction* and *velocity control*, $F(2, 38) = 6.3$, $p = .004$. Therefore, we analyzed the simple effects of the two *velocity control* conditions separately.

For the joystick conditions, a one-way ANOVA revealed a significant effect of *travel direction* on the traveled distance, $F(1.2, 23.4) = 9.4$, $p = .004$. The post-hoc test further showed that participants traveled significantly longer with the gaze-directed method than with the torso-directed $t(19) = 3.3$, $p = .011$, and virtual-body-directed method $t(19) = 3.0$, $p = .023$. The latter two did not differ $t(19) = 1.0$, $p = .927$.

For the leaning conditions, a one-way ANOVA also revealed

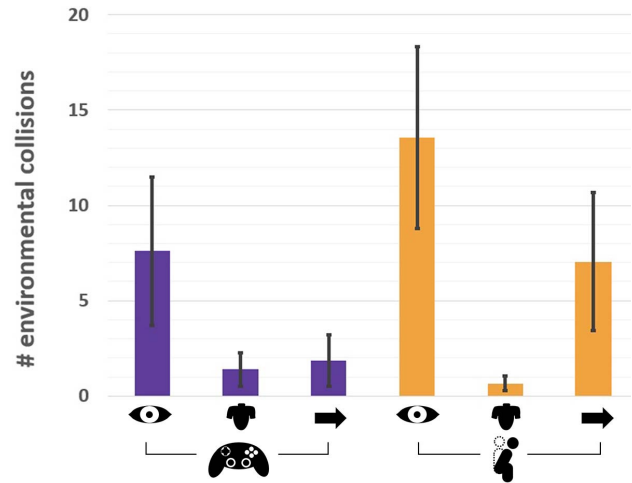


Figure 9: Mean number of collisions with the virtual environment for all conditions: {gamepad, leaning} $\times$ {gaze, torso, virtual-body-directed}. Error bars show the 95% confidence intervals. Refer to Section 5.3 for the significance analyses.

a significant effect of *travel direction* on the traveled distance, $F(2, 38) = 20.6$, $p < .001$. The post-hoc test further showed that participants traveled significantly longer with the gaze-directed method than with the torso-directed $t(19) = 6.1$, $p < .001$, and virtual-body-directed method $t(19) = 3.3$, $p = .011$. Additionally, the traveled distance was significantly longer when the virtual-body-directed method was used, compared to torso-directed leaning $t(19) = 3.3$, $p = .012$.

These simple effects, on the whole, can confirm (H3).

5.3 Environmental Collisions

The mean collisions participants had with the environment are given in Table 1 and depicted in Figure 9.

We hypothesized a main effect of *travel direction* on the number of collisions (H4) and thus performed a two-way ANOVA in this case. The statistical analysis revealed an interaction effect between the factors *travel direction* and *velocity control*, $F(1.4, 27.1) = 5.0$, $p = .023$. Therefore, we analyzed the simple effects of the two *velocity control* conditions separately.

For the joystick conditions, a one-way ANOVA revealed a significant effect of *travel direction* on the number of collisions, $F(1.1, 20.8) = 9.3$, $p = .005$. The post-hoc test further showed that participants collided significantly more frequently with the gaze-directed method than with the torso-directed $t(19) = 3.2$, $p = .013$, and virtual-body-directed method $t(19) = 2.9$, $p = .025$. The latter two did not differ $t(19) = -.9$, $p = 1.0$.

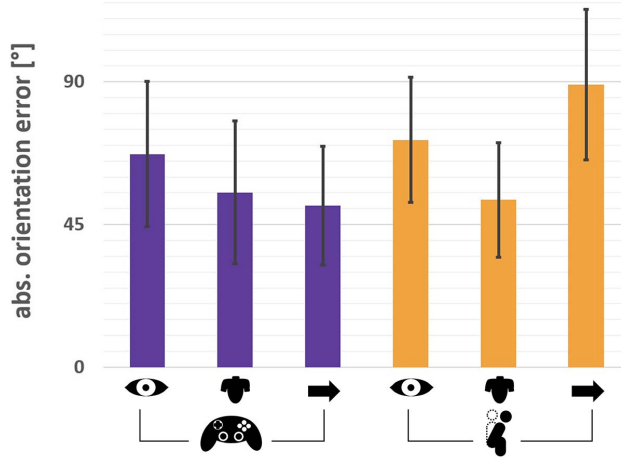


Figure 10: Mean absolute spatial orientation error for all conditions: {gamepad, leaning} $\times$ {gaze, torso, virtual-body-directed}. Error bars show the 95% confidence intervals. Refer to Section ?? for the significance analyses.

For the leaning conditions, a one-way ANOVA revealed a significant effect of *travel direction* on the number of collisions, $F(2, 38) = 16.5$, $p < .001$. The post-hoc test further showed that participants collided significantly more frequently with the gaze-directed method than with the torso-directed $t(19) = 5.5$, $p < .001$, and further showed a non-significant trend when compared to the virtual-body-directed method $t(19) = 2.63$, $p = .058$. Additionally, the number of collisions was significantly higher when the virtual-body-directed method was used, compared to torso-directed leaning $t(19) = 3.6$, $p = .006$.

These simple effects, on the whole, can confirm (H4) and (H4.2). However, (H4.1) cannot completely be confirmed, as a one-way ANOVA over all six methods showed significant differences regarding collisions, $F(2.5, 48.2) = 14.5$, $p < .001$, but the post-hoc tests did not show torso-directed leaning to significantly differ from torso-directed joystick $t(19) = -2.2$, $p = .2$, nor virtual-body-directed joystick $t(19) = -2.0$, $p = .285$, but only from gaze-directed joystick $t(19) = -3.7$, $p = .01$, gaze-directed leaning $t(19) = -5.5$, $p < .001$, and virtual-body-directed leaning $t(19) = -3.6$, $p = .01$, thus not being statistically significant the best but being with the best regarding number of collisions.

5.4 Spatial Orientation

After finishing a level, participants had to point to their starting point. The mean errors are given in Table 1 and depicted in Figure 10. A one-way ANOVA revealed no effect of the travel method on the spatial orientation error, $F(5, 95) = 2.0$, $p = .087$. Therefore, we cannot confirm (H5).

5.5 Usability

For all travel methods, participants answered the short version of the UEQ [18], giving an insight into the overall usability as well as into the pragmatic and hedonic quality of the interfaces. The mean UEQ-S scores are given in Table 1 and depicted in Figure 11. The pragmatic quality, for the most part, mirrors and thus validates the objective results.

For the hedonic quality, we performed a one-way ANOVA to confirm our hypotheses (H6). The statistical analysis revealed a significant effect of *travel direction* on the hedonic quality, $F(5, 95) = 10.0$, $p < .001$. Even though torso-directed leaning was the only method with a positive evaluation score ($> .8$), the post-hoc tests only

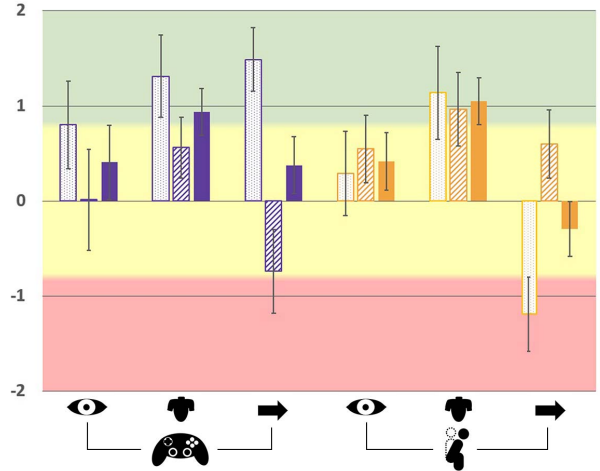


Figure 11: Mean UEQ-S scores [18] for all condition: {gamepad, leaning} $\times$ {gaze, torso, virtual-body-directed}. For each condition, the left bar depicts the pragmatic quality, the middle bar the hedonic quality, and the right one the overall score. The background divides positive $> .8$, neutral and negative $< .8$ evaluation scores [17]. Error bars show the 95% confidence intervals. Refer to Section 5.5 for the significance analyses.

showed the method to be significantly higher-rated as virtual-body-directed joystick $t(19) = 5.3$, $p = .001$. Thus (H6) can only partially be confirmed.

We also performed a one-way ANOVA on the impact of travel method to overall UEQ-S score and found a significant effect, $F(5, 95) = 10.7$, $p < .001$. The post-hoc test revealed statistically significant higher scores for torso-directed leaning, than for virtual-body-directed leaning $t(19) = 7.6$, $p < .001$. Additionally, there was a non-significant trend for a higher score, than virtual-body-directed joystick $t(19) = 3.1$, $p = .081$, and gaze-directed leaning $t(19) = 3.1$, $p = .087$. There was no difference found in comparison to the other methods.

Furthermore, virtual-body-directed leaning was rated significantly lower than torso-directed joystick $t(19) = -7.3$, $p < .001$, virtual-body-directed joystick $t(19) = -3.6$, $p = .030$, gaze-directed leaning $t(19) = -3.8$, $p = .019$, and torso-directed leaning $t(19) = -7.6$, $p < .001$. Only gaze-directed joystick was not significantly ranked lower $t(19) = -2.9$, $p = .143$.

5.6 Secondary Measures

Beyond the reported core measures and hypotheses, we measured some variables either very roughly to at least get a sketchy insight or for control reasons.

For example, we recorded forward and backward movement separately and have evaluated the first in Section 5.2. There was no difference in the number of backward movements between the travel methods, primarily due to the fact that there were almost no backward movements at all. During the training phases, however, participants were explicitly invited to test moving backward.

Then we asked participants if their well-being had changed after each method (see Section 4.4), while it was not possible to use a complete SSQ [9] here, due to time restrictions and the within-subjects design of the study. We ran a Friedman test that revealed a significant difference in reported well-being depending on which travel method was used before, $\chi^2(5) = 29.0$, $p < 0.01$. The post-hoc tests showed virtual-body-directed leaning having induced significantly more sickness than gaze-directed joystick $Z = 1.1$, $p = .013$, torso-directed joystick $Z = 3.8$, $p = .003$, and torso-directed leaning

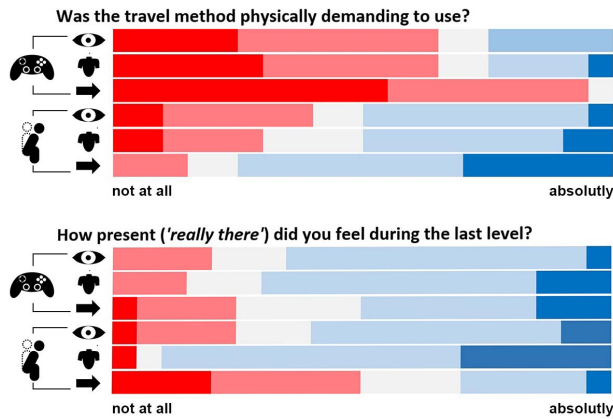


Figure 12: Reactions of all subjects to 5-point Likert scale items, regarding physical strain and presence for all conditions: {gamepad, leaning} $\times$ {gaze, torso, virtual-body-directed}.

$Z = 3.3, p = .017$.

Furthermore, we found indicators that the leaning-based methods felt more tiring, and torso-directed leaning induced the highest presence (see Figure 12), but we leave out the detailed statistics here.

6 DISCUSSION

Directly addressing the main research question, we could successfully show through all core measures that the ability to freely look around significantly supports users in relative motion (**H1**, **H3**), where the relative impact was much higher on time than on distance, which indicated that participants stopped their movement to look around, instead of taking many curvatures into account. The results also revealed that combining two functional interfaces, here leaning and the virtual-body-directed steering, do not necessarily form a new interface that is functional (**H2**).

We want to discuss the latter point a little bit further. The combination of leaning with virtual-body-directed steering was the only one that left margin in the form of implementation, i.e., there were other possibilities than putting the virtual rotation in conflict with the ability of looking around, e.g., on head roll or only induced by a physical rotation in or of the chair [11]. However, we had to decide on one, and related work had also shown that the interface works [21, 24]. We observed this in the training stage as well, where participants had no problems in steering around the pillars, but this did not require them to look around. Another factor was that the green arrow, which indicated the orientation of the virtual body, left the field of view when adjusting its orientation by looking to the side.

Furthermore, we found some evidence that using body cues for travel enhances the environmental presence, i.e., oneself accepting as being part of the virtual environment or accepting the surrounding environment, leading to fewer collisions with it (**H4**). However, we could not find a significant effect of body rotation and leaning on spatial orientation (**H5**). Nevertheless, we still believe that there could be an effect, but the task was very difficult, and thus, the variances high. This effect was even increased by the facts: that participants who had no idea were forced to guess, the participants' answers were not weighted by a self-reported ability to orient themselves, and we should have asked more often to gain more data. We want to apply these improvements in future experiments.

When evaluating the subjective usability, the gamepad methods have the tendency to be more task efficient and the leaning methods to feel better (see Figure 11), which is not surprising. However, it is interesting that the leaning methods objectively performed even

better as they were perceived and rated, which was also observed and mirrored in discussions with single participants after the experiment. Finally, a positive overall evaluation score was only reached by the two torso-directed methods.

Surprisingly, some participants complained about the leaning methods inducing more cybersickness, which would contradict the common sense that utilizing body cues do the opposite by increasing the self-motion perception. However, this neither was empirically confirmed with our minimal sickness measure nor led to worse performance in these cases.

Overall, the gamepad, as well as the leaning, performed comparable, which is interesting when considering that many more participants already had experiences with a gamepad. Especially, the two torso-directed methods performed well independently from the velocity control method. Additionally, the gamepad worked well with the virtual-body-metaphor, while its behavior should be very known to many users from 3D video games. Having all this in mind, the embodied methods additionally do not occupy the users' hands and integrate more naturally into the environment.

7 LIMITATIONS & FUTURE WORK

Next to the limitations that were already mentioned before, we would like to point out the following. While the experiment populations with computer science background often are not balanced in gender, this study was not completed by any female participant. This is important to note when drawing a conclusion to other populations.

In this study, we willingly tried and did isolate the effects of the ability to look around in an artificial and constructed task that should advantage methods with this ability. That does not necessarily mean that the results are reproducible in a more general task, where the ability to look around is less important. However, even when this ability is not always beneficial, it can contribute to a more natural interface and thus better experience. Still, the proposed interfaces should be validated in a more general task to exclude a negative impact on other activities, such as virtual-body-directed leaning in this study had.

Furthermore, in possible successors of this work, a pointing-directed technique should be compared to the torso-directed steering interfaces. Another interesting question is whether these results are transferable to standing scenarios as well, where it might be more exhausting to turn or lean physically. Last but not least, the impact on perception was just scratched in this work and should be further investigated.

8 CONCLUSION

In this work, we raised the question of whether the ability of freely looking around can enhance the performance in a combined relative motion and spatial orientation task. Therefore, we implemented a gaze and torso-directed steering method with upper body leaning, giving the velocity. We then investigated the impact of this freedom in a ground-based scenario and completed the conditions with a virtual body-directed method, then crossing all with device-based control conditions. In the conducted study, we found the torso-directed methods objectively performed the best and found torso-directed leaning subjectively rated the most usable one. In future work, we want to investigate if the results are transferable to standing scenarios and if torso-directed methods also perform in more general tasks.

REFERENCES

- [1] S. Bouchard, G. Robillard, J. St-Jacques, S. Dumoulin, M. Patry, and P. Renaud. Reliability and Validity of a Single-Item Measure of Presence in VR. *Proc. of IEEE Int. Workshop on Haptic, Audio and Visual Environments and Their Applications*, pp. 59 – 61, 2004.
- [2] D. A. Bowman, E. T. Davis, L. F. Hodges, and A. N. Badre. Maintaining Spatial Orientation During Travel in an Immersive Virtual

- Environment. *Presence: Teleoperators and Virtual Environments*, 8(6):618–631, 1999.
- [3] D. A. Bowman, D. Koller, and L. F. Hodges. Travel in Immersive Virtual Environments: An Evaluation of Viewpoint Motion Control Techniques. *Proc. of IEEE VR*, pp. 45–52, 1997.
 - [4] D. A. Bowman, D. Koller, and L. F. Hodges. A Methodology for the Evaluation of Travel Techniques for Immersive Virtual Environments. *Virtual reality*, 3(2):120–131, 1998.
 - [5] W. Chen, J. Chen, and R. H. Y. So. Visually Induced Motion Sickness: Effects of Translational Visual Motion Along Different Axes. *Contemporary Ergonomics and Human Factors (2011)*, pp. 281–287, 2011.
 - [6] C. Christou, A. Tzanavari, K. Herakleous, and C. Poullis. Navigation in Virtual Reality: Comparison of Gaze-Directed and Pointing Motion Control. *Proc. of IEEE Mediterranean Electrotechnical Conference*, pp. 1–6, 2016.
 - [7] T. Y. Grechkin and B. E. Riecke. Re-Evaluating Benefits of Body-Based Rotational Cues for Maintaining Orientation in Virtual Environments: Men Benefit From Real Rotations, Women Don't. *Proc. of Symposium on Applied Perception*, pp. 99–102, 2014.
 - [8] E. Guy, P. Punpongsonan, D. Iwai, K. Sato, T. Boubekeur, R. Abdrashitov, J. Yao, K. Singh, J.-M. Thierry, and J.-M. Thiery. LazyNav: 3D Ground Navigation with Non-Critical Body Parts. *Proc. of IEEE 3DUI*, pp. 43–50, 2015.
 - [9] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal. Simulator Sickness Questionnaire: An Enhanced Method for Quantifying Simulator Sickness. *Aviation Psychology*, 3(3):203–220, 1993.
 - [10] A. Kitson, A. M. Hashemian, E. R. Stepanova, E. Kruijff, and B. E. Riecke. Comparing Leaning-based Motion Cueing Interfaces for Virtual Reality Locomotion. *Proc. of IEEE 3DUI*, pp. 73–82, 2017.
 - [11] A. Kitson, B. E. Riecke, A. M. Hashemian, and C. Neustaedter. NaviChair: Evaluating an Embodied Interface Using a Pointing Task to Navigate Virtual Reality. *Proc. of ACM SUI*, pp. 123–126, 2015.
 - [12] E. Kruijff, B. Riecke, C. Trekowski, and A. Kitson. Upper Body Leaning can Affect Forward Self-Motion Perception in Virtual Environments. *Proc. of ACM SUI*, pp. 103–112, 2015.
 - [13] E. Langbehn, J. Wittig, N. Katzakis, and F. Steinicke. Turn Your Head Half Round: VR Rotation Techniques for Situations With Physically Limited Turning Angle. pp. 235–243, 2019.
 - [14] J. J. LaViola Jr, E. Kruijff, R. P. McMahan, D. Bowman, and I. P. Poupyrev. *3D User Interfaces: Theory and Practice*. Addison-Wesley Professional, 2017.
 - [15] T. Nguyen-Vo, B. E. Riecke, W. Stuerzlinger, D.-M. Pham, and E. Kruijff. NaviBoard and NaviChair: Limited Translation Combined with Full Rotation for Efficient Virtual Locomotion. *IEEE TVCG*, 2019.
 - [16] B. E. Riecke, B. Bodenheimer, T. P. McNamara, B. Williams, P. Peng, and D. Feureissen. Do We Need to Walk for Effective Virtual Reality Navigation? Physical Rotations Alone May Suffice. *Spatial Cognition VII*, pp. 234–247, 2010.
 - [17] M. Schrepp. User Experience Questionnaire Handbook. 2015.
 - [18] M. Schrepp, A. Hinderks, and J. Thomaschewski. Design and Evaluation of a Short Version of the User Experience Questionnaire (UEQ-S). *IJIMAI*, 4(6):103–108, 2017.
 - [19] E. A. Suma, S. L. Finkelstein, S. Clark, P. Goolkasian, and L. F. Hodges. Effects of Travel Technique and Gender on a Divided Attention Task in a Virtual Environment. *Proc. of IEEE 3DUI*, pp. 27–34, 2010.
 - [20] A. Von Kapri, T. Rick, and S. Feiner. Comparing Steering-Based Travel Techniques for Search Tasks in a Cave. *Proc. of IEEE VR*, pp. 91–94, 2011.
 - [21] D. Zielasko, S. Horn, S. Freitag, B. Weyers, and T. W. Kuhlen. Evaluation of Hands-Free HMD-Based Navigation Techniques for Immersive Data Analysis. *Proc. of IEEE 3DUI*, pp. 113–119, 2016.
 - [22] D. Zielasko, A. Meißner, S. Freitag, B. Weyers, and T. W. Kuhlen. Dynamic Field of View Reduction Related to Subjective Sickness Measures in an HMD-based Data Analysis Task. *Proc. of IEEE VR Workshop on Everyday Virtual Reality*, 2018.
 - [23] D. Zielasko, B. Weyers, M. Bellgardt, S. Pick, A. Meißner, T. Vierjahn, and T. W. Kuhlen. Remain Seated: Towards Fully-Immersive Desktop VR. *Proc. of IEEE VR Workshop on Everyday Virtual Reality*, pp. 1–6, 2017.
 - [24] D. Zielasko, B. Weyers, and T. W. Kuhlen. A Non-Stationary Office Desk Substitution for Desk-Based and HMD-Projected Virtual Reality. *Proc. of IEEE VR Workshop on Immersive Sickness Prevention*, 2019.